Inventory Reconciliation Report

Generated: June 11, 2026 at 04:05 UTC | Reconciliation ID: 13

Key Performance Indicators

Metric	Value
CSV Assets	15
Live (JSON) Assets	14
Missing Assets	0
Untracked Assets	3
Config Mismatches	1
Naming Mismatches	4
Total Discrepancies	8
Compliance Rate	100.0%

Executive Summary

AI generation error: 429 You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit.

* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash

Please retry in 8.709190433s. [links {

description: "Learn more about Gemini API quotas"

url: "https://ai.google.dev/gemini-api/docs/rate-limits"

}

, violations {

quota_metric: "generativelanguage.googleapis.com/generate_content_free_tier_requests"

quota_id: "GenerateRequestsPerMinutePerProjectPerModel-FreeTier"

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key: "model"

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}

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key: "location"

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quota_value: 5

}

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seconds: 8

}

]

AI Analysis

Inventory Validation Assessment

1. Data Quality Issues Detected:

* **Completeness:**

* **Untracked Assets:** 3 assets (21.4% of live infrastructure) are present in the live environment but not recorded in the CSV inventory. This represents a significant gap in inventory coverage, posing risks for security, compliance, and asset management.

* **Accuracy:**

* **Configuration Mismatches:** 1 asset (`PROD-DB-002`) has an inaccurate `ip_address` recorded in the CSV compared to its live state, indicating a direct data inaccuracy.

* **Naming Mismatches:** 4 discrepancies related to asset naming indicate inconsistencies between recorded and live asset identifiers. The specific detail "ID mismatch but name similar (ratio=1.00): CSV='' ~ Live=''" is highly concerning, suggesting either a failure to capture actual asset names or a flawed matching algorithm producing misleading similarity scores based on empty data.

* **Uniqueness & Consistency:**

* A critical issue is the detection of 4 `NAMING_MISMATCH` discrepancies where different `csv_asset_id`s (`FW-LON-03`, `MISSING-01`, `MISSING-02`, `MISSING-03`) are all being associated with the *same* `json_asset_id` (`PROD-DB-001`). This indicates a severe problem with data consistency, suggesting either duplicate entries in the CSV inventory for a single live asset or a highly unreliable matching process.

2. Duplicate or Missing Value Patterns:

* **Duplicate Association Pattern:** The most prominent pattern is the live asset `PROD-DB-001` appearing as the `json_asset_id` in 4 separate `NAMING_MISMATCH` entries. This strongly suggests that multiple, distinct CSV entries are erroneously mapped to a single live asset, implying either redundant data in the CSV or a fundamental flaw in the asset identification and matching process.

* **Missing Name Values:** The recurring detail "CSV='' ~ Live=''" within the `NAMING_MISMATCH` discrepancies indicates that the "name" fields, which are supposedly used for similarity comparison, are consistently empty. This is a critical pattern of missing data that undermines the validity of any "similarity ratio" calculation and points to a deeper issue in data collection or definition.

3. Data Integrity Concerns:

* **Referential Integrity Breach:** The one-to-many relationship (4 CSV assets to 1 live asset `PROD-DB-001`) observed in the naming mismatches directly compromises referential integrity. This makes it impossible to reliably map unique recorded assets to their live counterparts for a significant portion of the inventory.

* **Trustworthiness of Matching Logic:** The "ID mismatch but name similar (ratio=1.00): CSV='' ~ Live=''" detail raises serious doubts about the integrity and reliability of the asset matching algorithm. If similarity is based on empty strings, the entire matching process for these assets is questionable, potentially leading to widespread misidentification.

* **Auditability & Risk:** The combination of untracked assets and significant mapping discrepancies severely impacts the auditability of the IT infrastructure. It creates a high risk for compliance violations, security vulnerabilities (e.g., untracked rogue devices), and inefficient asset lifecycle management due to an unreliable inventory baseline.

4. Overall Data Quality Score (out of 100):

Score: 55/100

Justification:

The inventory data exhibits significant deficiencies across completeness, accuracy, and integrity dimensions.

* **Completeness:** 3 untracked assets (21.4% of live assets) represent a substantial gap in inventory coverage.

* **Accuracy:** 1 direct configuration mismatch and 4 naming mismatches point to inaccuracies.

* **Integrity:** The most severe issues are the 4 CSV assets incorrectly mapping to a single live asset (`PROD-DB-001`) and the empty name fields undermining the credibility of the "naming similarity" scores. These flaws severely compromise the reliability and trustworthiness of the inventory as a definitive record of the IT infrastructure. While there are no explicitly "missing" CSV assets, the quality of matching and recording for several assets is critically low.

Inventory Reconciliation Analysis

Reconciliation Summary:

* **CSV assets:** 15

* **Live assets:** 14

* **Missing (from CSV in Live):** 0

* **Untracked (Live not in CSV):** 3

* **Config mismatches:** 1

* **Naming mismatches:** 4

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1. Root Cause Analysis

* **Missing Assets (0):**

* **Likely Reason:** All assets listed in the CSV inventory were accounted for in the live infrastructure, either as a direct match, a naming mismatch, or a configuration mismatch. This indicates a good baseline for existing inventoried assets being present.

* **Untracked Assets (3):**

* **Likely Reasons:** These assets (`FW-LON-03-A`, `ROGUE-SW-99`, `ROGUE-SRV-88`) are present in the live environment but not recorded in the CSV inventory.

* **Lack of Integration/Automation:** New assets are being provisioned or deployed without being automatically registered in the inventory system.

* **Manual Process Failures:** If inventory updates are a manual process, there are likely missed steps or delays in adding new assets to the CSV.

* **Shadow IT/Unauthorized Deployment:** Assets could have been deployed outside of approved processes.

* **Misclassification:** Assets might exist but are classified differently in inventory, preventing a match, though the "untracked" status suggests no entry at all.

* **Config Mismatches (1):**

* **Likely Reason:** One asset (`PROD-DB-002`) shows a configuration discrepancy specifically on the `ip_address` field.

* **Configuration Drift:** Manual changes were made to the live asset's configuration (e.g., IP address reassignment) but were not updated in the inventory records.

* **Automated System Discrepancy:** Automated configuration tools might have made a change that wasn't propagated back to the inventory system.

* **Incorrect Initial Entry:** The IP address was incorrectly recorded in the CSV inventory from the start.

* **Naming Mismatches (4):**

* **Likely Reasons:** Four assets show discrepancies where the ID in CSV and live differ, but the reconciliation tool found them to be "similar" (ratio=1.00 based on some internal string comparison, not the raw IDs provided).

* **Inconsistent Naming Conventions:** Different naming standards are used between the inventory system and the live environment (e.g., `FW-LON-03` vs `PROD-DB-001`).

* **Renaming without Inventory Update:** Assets were renamed in the live environment, but the corresponding inventory records were not updated.

* **Typographical Errors:** Mistakes in asset IDs either in the CSV inventory or during live system setup.

* **Migration/Upgrade Issues:** Legacy naming conventions persist in one system while new ones are applied in another.

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2. Risk Assessment

Overall Risk Rating: Medium

Justification:

* **Untracked Assets (3):** The presence of untracked assets is a significant concern. These assets are entirely outside of documented control, posing security vulnerabilities (unpatched systems, unauthorized access), compliance risks, and operational blind spots. While the number is not excessively high (3 out of 14 live assets), it represents a notable portion of the live estate.

* **Config Mismatch (1):** A configuration mismatch, especially concerning an `ip_address`, can lead to network issues, service unavailability, or security policy violations. The severity is listed as "MEDIUM", indicating a potential but not immediate critical impact.

* **Naming Mismatches (4):** While generally less critical than untracked assets or config drift, pervasive naming inconsistencies can lead to operational inefficiencies, errors in reporting, difficulties during incident response, and challenges in automating asset management. The severity is "MEDIUM".

* **Missing Assets (0):** The absence of missing assets is a positive indicator, suggesting that the primary inventory records are largely reflected in the live environment.

The combined impact of untracked assets and configuration drift, even if individually rated medium, elevates the overall risk to a level requiring immediate attention to prevent future critical issues.

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3. Impact Analysis

* **Untracked Assets:**

* **Security:** Unauthorized access points, unpatched vulnerabilities, potential for malicious or forgotten systems, difficulty in forensic analysis.

* **Compliance:** Failure to meet regulatory requirements (e.g., SOX, GDPR, HIPAA) that mandate comprehensive asset inventories and configuration management. Audit failures.

* **Operations:** Lack of visibility, inability to properly allocate resources, increased troubleshooting time for unknown devices, potential for network conflicts.

* **Financial:** Mismanagement of licenses, incorrect depreciation, unexpected costs associated with undocumented assets.

* **Config Mismatches:**

* **Service Availability:** Incorrect IP addresses can lead to network connectivity issues, service outages, or routing problems.

* **Security:** Deviations from security baselines (e.g., incorrect firewall rules, unapproved services) can create exploitable vulnerabilities.

* **Compliance:** Failure to adhere to security policies or industry standards, leading to audit findings.

* **Troubleshooting:** Increased time and complexity in diagnosing issues due to inaccurate configuration data in the inventory.

* **Naming Mismatches:**

* **Asset Management Inefficiency:** Difficulty in locating, identifying, and managing specific assets, leading to delays in maintenance, upgrades, and incident response.

* **Data Integrity:** Inaccurate inventory data undermines the reliability of asset management systems, reporting, and decision-making.

* **Automation Failures:** Automated scripts or tools relying on consistent naming conventions may fail, requiring manual intervention.

* **Audit Complications:** Challenges in reconciling records during audits, potentially leading to findings or fines.

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4. Trend Indicators

* **Systemic Issue with Asset Provisioning and Inventory Updates:** The presence of **3 untracked assets** suggests a consistent failure in the process of onboarding new assets and updating the inventory. This is unlikely to be an isolated incident, but rather indicates a gap in automation or manual procedures.

* **Configuration Drift Management Weakness:** The **1 config mismatch** on an `ip_address` suggests that changes are occurring in the live environment without being accurately reflected in the inventory. This could point to a lack of robust configuration management (CMDB) practices, automated reconciliation, or change control. While only one instance, it highlights a potential for broader configuration drift.

* **Inconsistent Naming Practices:** The **4 naming mismatches** indicate a potential lack of strict naming conventions enforcement or a process where asset names/IDs are changed in one system but not synchronized with the inventory. This is a common systemic issue in larger environments.

* **Overall Discrepancy Rate:** With 8 total discrepancies (3 untracked, 1 config, 4 naming) across a relatively small base of 14 live assets (or 15 CSV assets), the reconciliation rate is not ideal. This pattern, particularly across multiple categories, points towards **systemic weaknesses** in asset lifecycle management processes, configuration management, and inventory synchronization rather than isolated errors. There's no indication of one-off human error; instead, the types of discrepancies suggest process-level gaps.

Discrepancy Details

Type	CSV Asset	JSON Asset	Severity	Details
CONFIG MISMATCH	PROD-DB-002	PROD-DB-002	MEDIUM	Configuration mismatch on fields: ip_address

NAMING MISMATCH	FW-LON-03	PROD-DB-001	MEDIUM	ID mismatch but name similar (ratio=1.00): CSV='' ~ Live=''
NAMING MISMATCH	MISSING-01	PROD-DB-001	MEDIUM	ID mismatch but name similar (ratio=1.00): CSV='' ~ Live=''
NAMING MISMATCH	MISSING-02	PROD-DB-001	MEDIUM	ID mismatch but name similar (ratio=1.00): CSV='' ~ Live=''
NAMING MISMATCH	MISSING-03	PROD-DB-001	MEDIUM	ID mismatch but name similar (ratio=1.00): CSV='' ~ Live=''
UNTRACKED ASSET	—	FW-LON-03-A	MEDIUM	Asset 'FW-LON-03-A' found in live infrastructure but not in
UNTRACKED ASSET	—	ROGUE-SW-99	MEDIUM	Asset 'ROGUE-SW-99' found in live infrastructure but not in
UNTRACKED ASSET	—	ROGUE-SRV-88	MEDIUM	Asset 'ROGUE-SRV-88' found in live infrastructure but not in

Recommendations

• AI generation error: 429 You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit.

• * Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash

• Please retry in 9.611063174s. [links {

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• url: "https://ai.google.dev/gemini-api/docs/rate-limits"

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• key: "location"

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